Smart Hospital Management System (SHMS) - High-Level System Architecture

1. Presentation & Client Touchpoints Layer

- **Patient Portal & Mobile App**
  (Appointments, EHR View, Teleconsult)
  [React Native / Flutter]
- **Doctor & Nursing Station App**
  (Clinical Notes, E-Prescriptions, Orders)
  [Web / Tablet Workstation]
- **Administration & Operations Desk**
  (Billing, Pharmacy, Bed & Staff Mgmt)
  [Enterprise Web Dashboard]
- **Smart Medical IoT & Wearables**
  (ICU Bed Monitors, SpO2, ECG, Infusion)
  [Embedded Firmware / Bluetooth]
- **Emergency & Ambulance Units**
  (Live GPS Tracking & Patient Telemetry)
  [Rugged Tablets / 5G Link]

HTTPS / TLS 1.3 / REST
mTLS / REST
MQTT / CoAP / 5G Telemetry
Emergency GPS Telemetry

2. Gateway, Traffic Routing & Security Layer

- **Unified API Gateway / Reverse Proxy**
  (Rate Limiting, SSL Termination, Load Balancer)
  [Kong / NGINX / Ocelot]
- **Authentication & Identity Management**
  (OAuth 2.0 / OpenID / JWT / HIPAA RBAC)
  [Keycloak / Auth0]
- **IoT Edge Message Broker**
  (Protocols: MQTT / CoAP / WebSockets)
  [EMQX / Mosquitto / Kafka Connector]

High-Throughput Vitals Ingestion
gRPC / Routed REST Requests

3. Application & Microservices Layer (Core Hospital Business Logic)

- **Electronic Health Record**
  (EHR / Medical History / HL7 FHIR)
- **Appointments & Scheduling**
  (OPD Queues, Doctor Rosters)
- **Pharmacy & Inventory**
  (Stock Level Alerts & Batch Track)
- **Billing, Insurance & Claims**
  (Invoice Gen, TPA Pre-auth)
- **Laboratory & Diagnostics**
  (Pathology, Radiology, LIS Reports)
- **Real-Time IoT Telemetry Engine**
  (Vitals Stream Processing & ICU Triggers)
- **AI Diagnostics & Decision Support**
  (Predictive Triage, Anomaly ML Models)
- **Notification & Alert Dispatcher**
  (Critical Code Alarms, SMS, Push, Email)

Secure Webhooks / APIs
Session Caching & Queues
FHIR Claims Integration
ACID Relational Queries
DICOM Blob Store & Fetch
High-Speed Time-Series Ingestion

4. Data Persistence & Storage Layer

- **Relational Database**
  (PostgreSQL / Enterprise DB)
  • Patient Master Records
  • Staff & Doctors Directory
  • Financials & Billing Data
- **Time-Series & Document DB**
  (TimescaleDB / Influx / MongoDB)
  • Real-time Sp02, Heart Rates
  • Unstructured Clinical Notes
  • Wearable Telemetry Streams
- **In-Memory Cache & Queue**
  (Redis Cluster / Ra
  • Active User Sessions
  • Live OPD Token Queues
  • Event Bus Broker
- **Cloud Object Store / PACS**
  (MinIO / AWS S3 Blob Store)
  • High-Res DICOM Scans
  • MRI, CT-Scan & X-Ray Imaging
  • Lab Diagnostic PDFs

Push Notifications / SMS Triggers

5. External Integrations & Interoperability Layer

- **Payment Gateways**
  (Stripe / Razorpay / NetBanking)
  Online Invoicing & Copay
- **Insurance TPAs & Clearinghouses**
  (FHIR / HL7 v2/v3 Standards)
  Automated Cashless Claims
- **Omnichannel Comm Channels**
  (Twilio, AWS SNS, SendGrid)
  Urgent SMS, WhatsApp, Voice Calls
- **National Digital Health Registry**
  (ABHA / ABDM / Govt EHR Network)
  Inter-Hospital Health Records